

★ WEEK 6 · PART 2 of 3

Granger Causality

Does the Past of X Help Predict Y?

The predictive test · four outcomes · why it is NOT real causality

A fitted VAR lets us ask a sharp question: does one series' history improve the forecast of another? That is Granger causality — powerful, useful, and dangerously easy to misread.

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PART 2 · GRANGER CAUSALITY

Predictive, Not Causal

The Granger test, its four possible outcomes, the crucial warning about what it does and does not mean, and its honest use for lead-lag analysis in markets.

From Co-Movement to a Question

The VAR sets up a testable relationship

1

The VAR estimated the cross-terms

Part 1 gave us off-diagonal coefficients linking each series to the other's past. Now we ask whether those links are statistically real.

2

The question is about prediction

Does knowing X's history let us forecast Y better than knowing Y's history alone? If so, X 'Granger-causes' Y.

3

The answer is easy to over-interpret

A significant test is about predictive content, not cause and effect. Getting this distinction right is the heart of Part 2.

Part 2 in one line: Granger causality tests prediction, and prediction is not causation.

The Core Question

Does X's past improve the forecast of Y?

Granger's insight: if the past of X helps predict Y beyond what Y's own past already provides, then X carries useful information about Y's future. That is all the test really claims.

The intuition

- Start with the best forecast of Y from its own history.
- Now add X's history. Does the forecast improve?
- If yes, X 'Granger-causes' Y — it leads Y in a predictive sense.

Why it is testable

- It reduces to comparing two forecasts' errors.
- That comparison is a standard statistical test.
- No hidden theory — just predictive value, measured.

Granger Causality, Defined

A statement about forecast error

$$X \text{ Granger-causes } Y \iff \text{MSE}(Y_t | Y_{<t}, X_{<t}) < \text{MSE}(Y_t | Y_{<t})$$

Reading it

- The left side conditions on both series' past.
- The right side conditions on Y's past alone.
- If adding X strictly lowers the forecast error, X Granger-causes Y.

What it is really saying

- 'Cause' here means 'helps predict', nothing more.
- It is defined entirely in terms of forecast accuracy.
- The name is historical — and, we'll see, misleading.

Restricted vs Unrestricted

An F-test on the cross-lag coefficients

$$\underbrace{Y_t = c + \sum a_i Y_{t-i} + \sum b_i X_{t-i} + \varepsilon_t}_{\text{unrestricted}} \xrightarrow{F\text{-test}} H_0 : b_1 = \dots = b_q = 0$$

The mechanics

- Fit Y on its own lags (restricted).
- Fit Y on its own lags PLUS X's lags (unrestricted).
- F-test whether X's lag coefficients are jointly zero.

The verdict

- Reject the null $\rightarrow$ X's past adds predictive value.
- Fail to reject $\rightarrow$ X adds nothing beyond Y's own past.
- It is a joint test on all of X's lag terms at once.

Does X's Past Add Anything?

Forecast Y from its own past, then again with X's past added. If the error drops significantly, X Granger-causes Y — the whole test in one picture.

The Test: Does X's Past Add Anything?

Forecast Y two ways and compare. If adding X's past cuts the error, X Granger-causes Y.

Restricted model

predict Y from its OWN past only

$$Y_t = c + a_1 Y_{(t-1)} + \dots + e_t$$

gives a forecast error:

MSE (restricted)

Unrestricted model

add the past of X as well

$$Y_t = c + a Y_{(t-1)} + b X_{(t-1)} + e_t$$

gives a smaller error?

MSE (unrestricted)

The F-test asks: is the drop in error statistically real?

If yes (the b coefficients are jointly non-zero) → X Granger-causes Y. If no → X adds no predictive value.

Running the Test

Five steps to a clean Granger result

1 · Stationarize

both series

Difference so X and Y are stationary (Part 1).

2 · Pick lags

order p

Use the VAR lag order; same p for both.

3 · Two fits

restrict?

Estimate the restricted and unrestricted models.

4 · F-test

jointly zero?

Test X's lag coefficients jointly against zero.

5 · Both ways

direction

Repeat with roles swapped to get the direction.

Always test BOTH directions — $X \rightarrow Y$ and $Y \rightarrow X$ are separate questions with separate answers.

Four Possible Outcomes

A pair relates in one of four ways

One-directional

- X Granger-causes Y but not the reverse — X leads Y.
- Or Y Granger-causes X — Y leads X.
- A clear lead-lag ordering in the predictive sense.

Two-directional or none

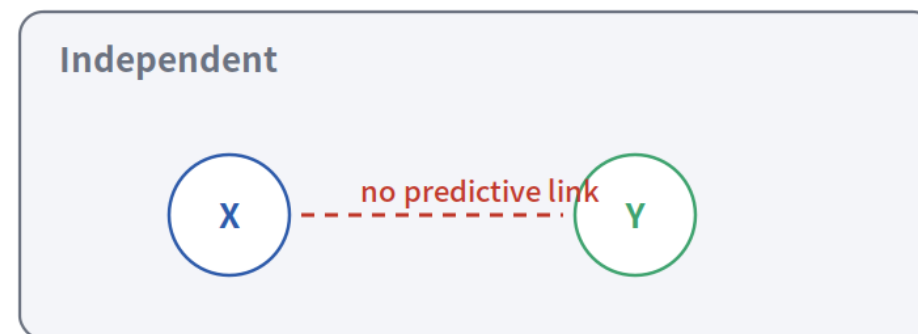
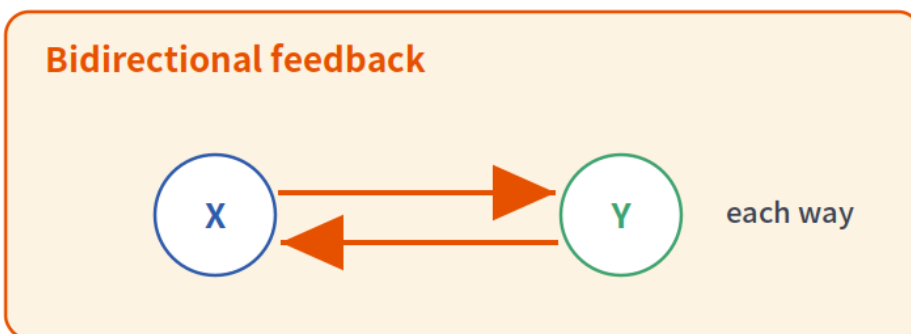
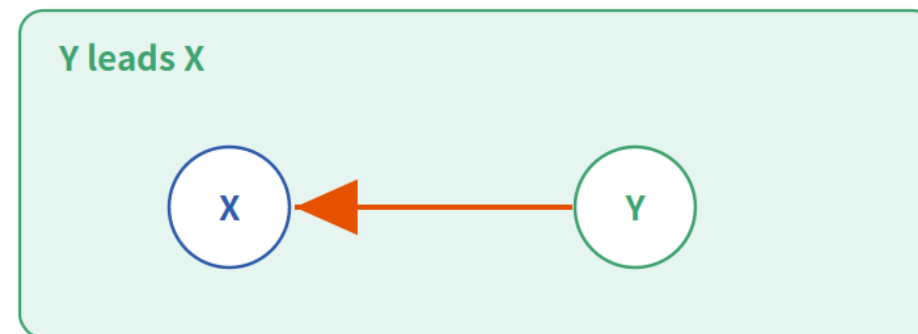
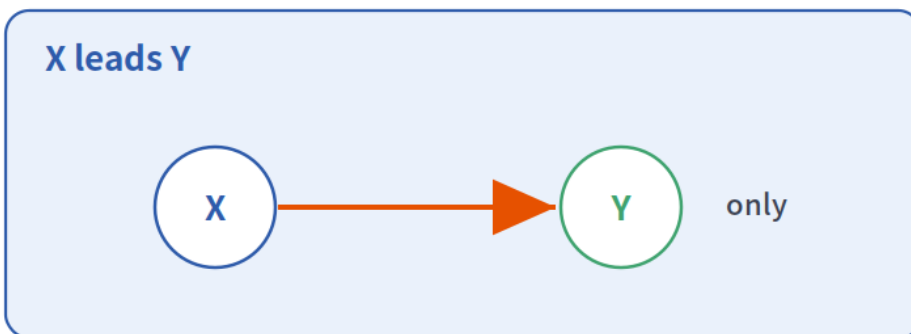
- Both directions significant — a feedback loop.
- Neither significant — no predictive link between them.
- Each case tells a different story about the pair.

Directions of Granger Causality

X leads Y, Y leads X, mutual feedback, or independence. Testing both directions places every pair into exactly one of these four boxes.

Four Possible Outcomes

A pair can relate in one of four predictive ways.



Granger Causality Is NOT Real Causality

A name that has misled generations

Granger causality means 'X's past helps predict Y'. It does NOT mean X makes Y happen. A predictive link can arise with no causal mechanism at all — the name promises far more than the test delivers.

What it DOES establish

- That X's history contains information about Y's future.
- A directional, time-ordered predictive relationship.
- A useful signal for forecasting and screening.

What it does NOT establish

- That intervening on X would change Y.
- That there is any causal mechanism at all.
- That a hidden third variable isn't driving both.

Why the Name Misleads

Three ways prediction diverges from cause

Common culprits

- A confounder Z drives both X and Y at different lags.
- Expectations: markets price the future, reversing apparent lead-lag.
- Timing artifacts from sampling or reporting delays.

The honest reading

- 'X Granger-causes Y' = 'X's past predicts Y'. Full stop.
- Add 'in a purely statistical, non-causal sense' in your head.
- Reserve causal claims for experiments or careful design.

A Cautionary Tale

When prediction has no causal content

A falling barometer 'Granger-causes' storms: its past predicts them. But smashing the barometer will not stop the weather — a hidden driver (air pressure) moves both.

The structure

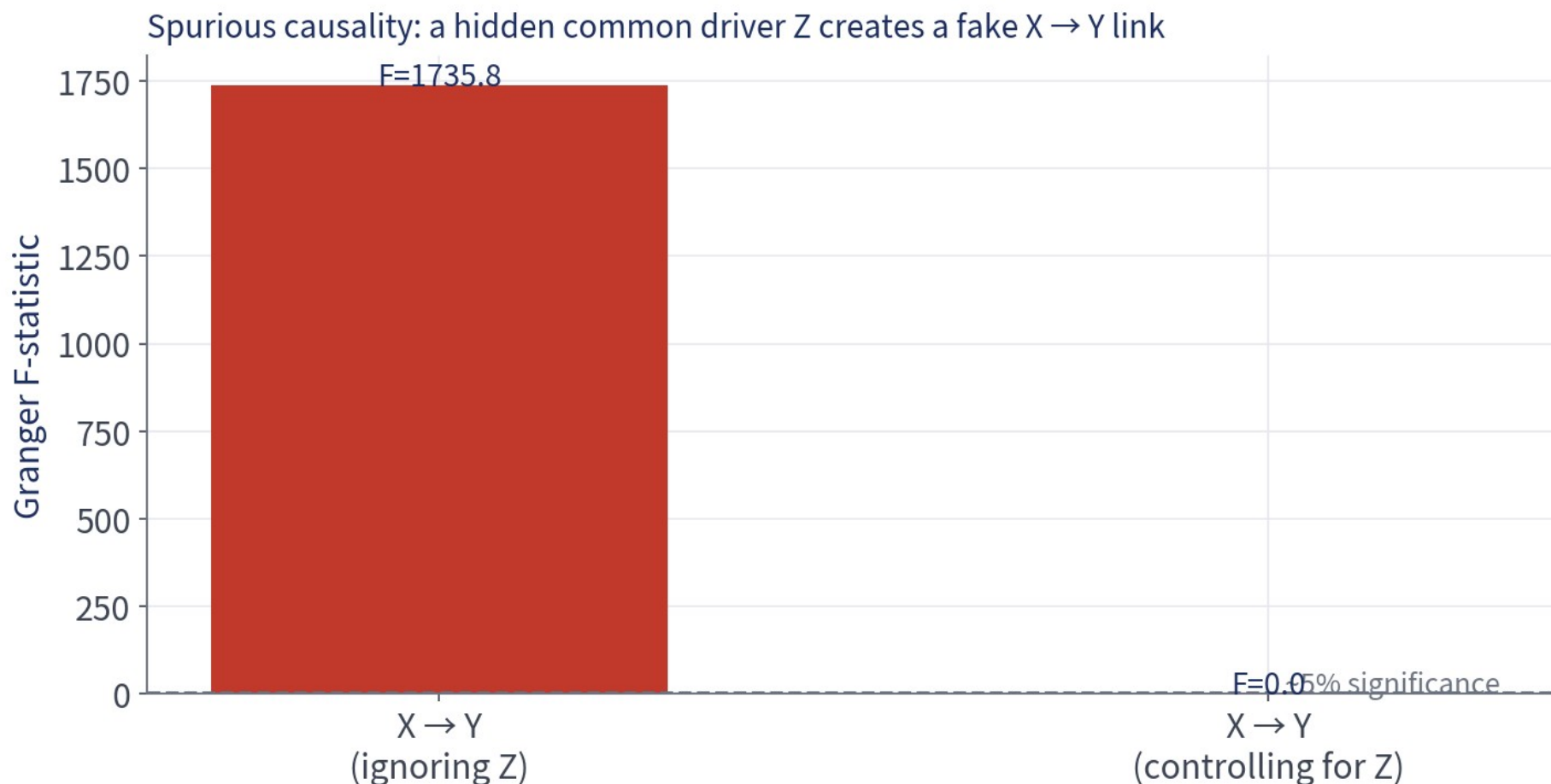
- A common cause (pressure) drives both barometer and storm.
- The barometer reacts first, so its past predicts the storm.
- The test correctly finds predictive content — and nothing more.

The finance version

- One asset may lead another only because both track a factor.
- The 'leader' is just quicker to reflect shared news.
- Trading on it as a cause can fail the moment the factor shifts.

Spurious Causality From a Confounder

X appears to strongly Granger-cause Y — until you control for the hidden common driver Z, and the effect collapses. Predictive links can be entirely indirect.



Instantaneous Causality

When two series move in the SAME period

instantaneous causality $\iff \text{Cov}(\varepsilon_{1,t}, \varepsilon_{2,t}) \neq 0$ (contemporaneous)

What it is

- Granger causality is about LAGGED prediction.
- But shocks can also be correlated within the same period.
- That contemporaneous link is 'instantaneous causality'.

Why it matters

- It shows up as correlated VAR residuals.
- It has no direction — it is symmetric by nature.
- It is exactly what forces us to orthogonalize in Part 3.

Beyond Pairs: Block Exogeneity

Granger causality in a larger system

The generalization

- With more than two variables, test a GROUP of lags jointly.
- Does a block of variables help predict another?
- This is 'block exogeneity' — the multivariate Granger test.

Why condition on others

- Testing $X \rightarrow Y$ alone can be fooled by a third variable.
- Including it can make a spurious link disappear.
- Always ask: what else should be in the system?

Lead-Lag in Markets

Where Granger causality earns its keep

Used honestly — as a forecasting and screening tool, not a causal claim — Granger causality reveals which series carry early information about others: the essence of lead-lag analysis.

Classic lead-lag patterns

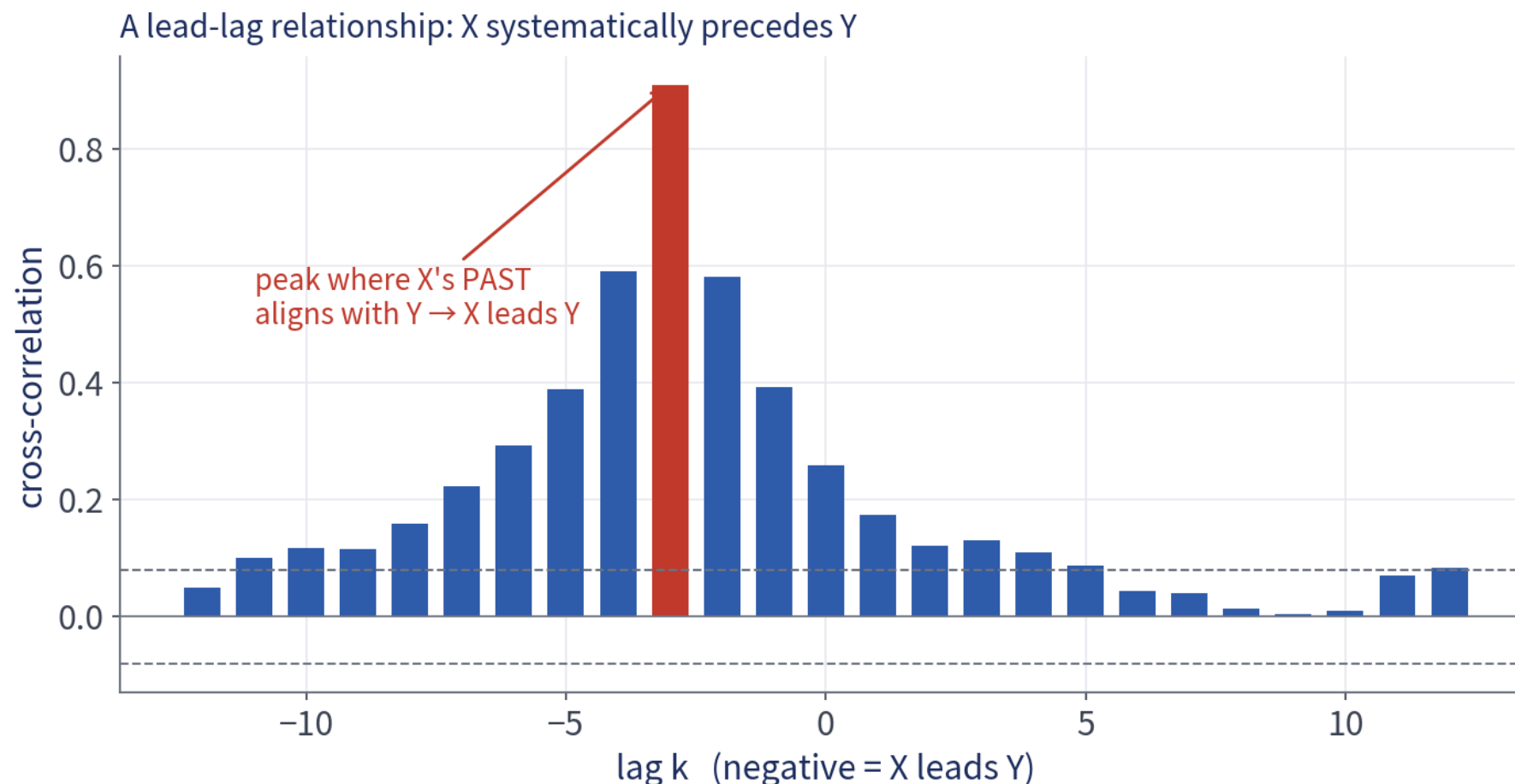
- Futures often lead spot prices.
- Large, liquid stocks can lead smaller, related ones.
- Some macro releases lead sector returns.

How to use it

- As a screen for candidate predictors, then validate out of sample.
- As a description of information flow, not a trading guarantee.
- Combine with economic reasoning — never rely on it alone.

A Lead-Lag Relationship

The cross-correlation peaks where X's past aligns with Y's present: X systematically precedes Y. That predictive lead is what a Granger test formalizes.



Correlation, Granger, Causation

Three rungs on the same ladder

Property	Correlation	Granger causality	True causation
Uses time order	no	yes	yes
Predictive	weak / static	yes, by definition	yes
Rules out confounders	no	no	yes (with design)
What it licenses	association	forecasting	intervention

Granger sits between correlation and causation: it adds time and prediction, but not the right to intervene.

What a Significant Result Tells You

And, just as important, what it doesn't

It DOES tell you

- X's past has predictive value for Y in this sample.
- There is a directional, time-ordered statistical link.
- X is a candidate input for forecasting Y.

It does NOT tell you

- That X causes Y in any mechanical sense.
- That the link will hold out of sample or persist.
- That no omitted variable explains it away.

Granger Causality and Efficient Markets

Why strong predictability is rare and fragile

The efficiency view

- If X reliably predicted Y's returns, traders would exploit it.
- Doing so tends to erase the very pattern (Week 1).
- So robust, tradable Granger causality in returns is scarce.

Where it does appear

- At high frequency, as information propagates between assets.
- In volumes, volatility, and fundamentals more than raw returns.
- As transient lead-lag that decays as it becomes known.

Using Granger Causality Well

A disciplined checklist

Do

- Treat it as a forecasting and screening tool.
- Test both directions and condition on relevant variables.
- Validate any predictor out of sample before trusting it.

Don't

- Don't read it as cause and effect.
- Don't ignore possible confounders or expectations.
- Don't assume a sample result will persist.

Test Granger Causality

Direct Claude Code through the test — and its caveats

GOAL Test Granger causality both ways on a pair, then show how a confounder can create a spurious result.

[INPUT] Prompt · Claude Code

Write and run `granger_lab.py` that:

1. takes two stationary series and runs the Granger test in BOTH directions with `statsmodels`, reporting the F-stats and p-values by lag;
2. plots their cross-correlation to show any lead-lag;
3. then simulates a confounder `Z` driving both `X` and `Y` at different lags, and shows the spurious `X→Y` result vanishing once `Z` is controlled for.

Interpret each result carefully.

How to work with it

- Report BOTH directions — they can differ.
- The CCF peak location shows who leads whom.
- Watch the spurious link die when `Z` is added.
- Never phrase the output as 'X causes Y'.

Deliverable: `granger_lab.py` + a both-directions result and a confounder demonstration.

Three Questions Before You Move On

If you can answer these, Part 2 has landed

Q1	The test says X Granger-causes Y. Can you conclude X causes Y?	No — prediction only
Q2	Both $X \rightarrow Y$ and $Y \rightarrow X$ are significant. What is this called?	Bidirectional feedback
Q3	A significant link vanishes when you add variable Z. What happened?	Z was a confounder

All three come from the meaning of the test and the confounder problem.

Three Misconceptions to Avoid

The errors this whole part guards against

Common myths

- "Granger causality proves X causes Y."
- "One direction is enough to characterize a pair."
- "A significant in-sample result will make money."

The reality

- It proves predictive content only — never a mechanism.
- Test both directions; feedback and independence are distinct.
- Validate out of sample; efficient markets erase easy edges.

Three Things to Carry Forward

Part 2 in one screen

1

Granger causality is a prediction test

It asks whether X's past improves the forecast of Y — an F-test comparing a restricted and an unrestricted model, run in both directions.

2

It is not real causality

Confounders, expectations, and timing can create predictive links with no mechanism. Read every result as 'helps predict', never 'causes'.

3

Used honestly, it maps information flow

As a screening and lead-lag tool — validated out of sample and conditioned on the right variables — it reveals which series carry early information.

Next — Part 3: correlated shocks, orthogonalization, and tracing shocks with IRF and FEVD.